

MURORAN YAKUMO, Japan

WMO: 474230

Lat: 42.3117N

Lon: 140.9753E

Elev: 163

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.2	20.2	7.2	8.0	20.1	9.6	9.0	21.8	33.3	26.4	30.4	26.0	15.3	300	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	79.6	73.0	77.0	71.5	74.9	70.5	74.5	77.8	73.0	75.7	71.5	73.9	8.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	124.8	76.6	72.0	118.9	74.9	70.6	113.5	73.5	38.2	78.1	36.9	75.9	35.5	73.9	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.9	24.6	21.5	15.8	84.0	2.5	2.6	14.0	85.8	12.6	87.3	11.1	88.8	9.3	90.6		
			13.9	76.3	2.6	1.9	12.0	77.6	10.5	78.7	9.1	79.8	7.2	81.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	28.5	29.1	34.7	43.4	51.8	58.5	66.1	70.1	65.6	55.2	43.3	32.9
	DBStd	15.27	4.63	4.85	4.98	4.94	5.09	4.89	4.96	4.20	4.37	5.22	6.92	5.69
	HDD50	2736	666	585	475	207	38	2	0	0	0	16	217	530
	HDD65	6380	1131	1005	940	648	409	202	47	7	43	303	650	995
	CDD50	2149	0	0	0	9	95	258	499	623	469	178	18	0
	CDD65	316	0	0	0	0	0	8	81	165	62	0	0	0
	CDH74	668	0	0	0	0	2	11	174	402	79	0	0	0
	CDH80	57	0	0	0	0	0	0	11	43	3	0	0	0
Wind	WSAvg	10.3	12.8	11.8	11.0	9.8	9.2	8.1	7.9	7.6	8.5	10.4	12.4	13.6
Precipitation	PrecAvg	48.1	2.9	2.2	2.1	2.8	4.0	3.7	5.8	6.9	6.4	4.0	4.0	3.4
	PrecMax	58.3	5.1	5.1	3.9	5.1	6.7	9.3	12.5	13.9	12.6	5.8	10.7	5.9
	PrecMin	39.8	1.6	1.0	0.7	0.6	1.3	1.1	1.5	2.7	1.3	2.0	2.4	2.0
	PrecStd	5.2	0.9	1.0	0.8	1.3	1.5	1.9	2.5	3.0	2.8	1.1	1.7	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	43.0	51.4	62.7	71.2	75.5	81.5	84.0	79.6	68.9	60.9	51.5
		MCWB	39.7	39.3	44.1	51.6	57.6	66.2	73.1	75.7	72.9	62.5	55.7	48.9
	2%	DB	39.5	40.0	47.1	57.8	66.5	71.8	78.3	80.8	76.1	66.3	58.1	47.4
		MCWB	36.6	36.5	41.5	48.8	56.0	63.9	71.6	74.2	70.6	61.0	53.8	44.6
	5%	DB	37.2	37.9	44.1	53.8	63.0	68.6	75.7	78.1	73.9	64.4	55.8	44.1
		MCWB	34.1	34.6	39.6	46.3	54.5	62.2	70.3	72.9	69.2	59.8	51.8	40.6
	10%	DB	35.2	36.1	41.7	50.7	59.5	65.8	73.2	75.9	71.9	62.7	53.6	41.5
		MCWB	32.1	33.0	37.7	44.5	52.8	61.0	69.2	71.7	67.8	58.2	49.7	37.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	40.4	45.5	53.3	60.4	67.9	74.7	77.1	74.7	65.4	57.5	49.2
		MCDB	41.9	42.0	49.6	60.3	68.2	73.7	79.0	82.0	77.2	66.8	59.5	50.9
	2%	WB	37.2	37.3	42.6	49.7	57.5	65.3	73.0	75.3	72.6	62.6	54.7	44.9
		MCDB	39.1	39.2	46.1	56.5	63.9	69.9	76.8	79.2	74.7	64.8	57.2	47.1
	5%	WB	34.7	35.2	40.2	47.4	55.6	63.3	71.3	73.9	70.8	60.9	52.4	41.1
		MCDB	36.7	37.3	43.4	52.1	61.0	67.3	74.5	76.8	72.7	63.5	55.3	43.8
	10%	WB	32.4	33.4	38.2	45.7	53.8	61.6	69.6	72.6	68.8	59.1	50.0	38.1
		MCDB	35.0	35.8	41.1	49.6	58.1	65.0	72.4	75.0	71.0	62.0	53.4	41.2
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.8	9.2	10.9	11.2	9.5	8.2	7.8	9.2	10.0	9.2	7.4
		MCDBR	9.2	9.6	12.2	15.5	17.2	14.2	12.5	11.2	10.5	10.6	11.6	10.9
		MCWBR	8.5	8.5	9.1	9.1	9.5	8.1	6.9	6.3	6.8	8.0	10.4	10.8
	5% WB	MCDBR	9.5	9.6	11.9	13.9	14.9	12.4	11.0	9.9	8.5	9.3	11.4	11.0
		MCWBR	9.7	9.3	9.8	9.2	9.1	7.7	6.6	5.9	6.2	8.0	11.2	11.7
Clear-Sky Solar Irradiance	taub	0.346	0.395	0.469	0.528	0.548	0.523	0.513	0.481	0.422	0.407	0.389	0.363	
	taud	2.102	1.962	1.849	1.794	1.839	1.986	2.085	2.192	2.310	2.209	2.152	2.091	
	Ebn at Noon	240	245	241	237	237	244	244	249	256	241	223	220	
	Edh at Noon	34	46	58	66	65	56	50	44	36	36	32	31	
All-Sky Solar Radiation	RadAvg	453	690	1072	1430	1592	1530	1350	1350	1312	1162	868	516	373
	RadStd	27	56	72	151	144	135	137	111	113	53	46	27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page